

Assessing TARP

Dinara Bayazitova

Kenan-Flagler Business School, University of North Carolina

Anil Shivdasani

Kenan-Flagler Business School, University of North Carolina

We study the government equity infusions into banks under the Capital Purchase Program (CPP) of the Troubled Asset Relief Program (TARP). Strong banks opted out of CPP, and equity infusions were provided to banks that posed systemic risk and faced high financial distress costs but had strong asset quality. Concerns over executive compensation led banks to reject CPP infusions and exit the program. CPP infusions did not have meaningful certification effects, but the subsequent stress tests conducted for the major banks had significant certification effects. CPP equity infusions increased investor expectations regarding future regulatory interventions in the banking sector. (*JEL* G18, G21, G28)

1. Introduction

The Troubled Asset Relief Program (TARP) was the largest government bailout in U.S. history. A cornerstone of this program was the government's purchase of equity in financial institutions. In this article, we examine the equity infusions made under the Capital Purchase Program (CPP) and the Capital Assistance Program (CAP) of TARP. Under CPP, banks could apply for preferred stock capital infusions from the U.S. Treasury, while CAP mandated stress tests for the largest banks, requiring those with capital shortfalls to raise common equity. Several observers have questioned whether these equity infusions were a cost-efficient solution and if alternative strategies would have yielded better outcomes (Hoshi and Kashyap 2010; Diamond and Rajan 2010; Veronesi and Zingales 2010). Our goal is to provide systematic evidence on the equity injections and the incentives of banks to participate in these capital infusions.

During a financial crisis, capital levels of banks are depleted and raising new capital in public markets is difficult. A government capital injection program can stabilize banks by providing a source of capital when public

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market alternatives are unavailable. Capital injections can also signal the government's favorable private information about the bank's financial health, thereby lowering information asymmetry and making it easier to raise capital in public markets. However, by propping up weak institutions with illiquid assets, capital injections increase the specter of asset fire sales and can cause liquidity to dry up in the financial system (Diamond and Rajan 2009, 2010). In addition, a government intervention can increase the market's expectation of future interventions and regulatory seizures, potentially impeding financial sector recovery.

The incentives of banks to participate in a capital injection program are an important issue. Based on the Japanese experience, Hoshi and Kashyap (2010) argue that the success of a financial rescue program depends critically on the willingness of weak banks to participate in it. Hoshi and Kashyap (2010) note that a bank may refuse government assistance if the capital injection generates an adverse signal that the bank is expected to have high future losses. In addition, if the capital infusion is senior to common equity holders, as was the case with the preferred stock infusions under CPP, the upside for shareholders in the event of recovery is reduced. Thus, because of debt overhang, much of the recovery in firm value accrues to debt holders, leaving stockholders with little incentive to participate in the rescue. In addition to these signaling and debt overhang obstacles, we argue that the incentives of banks to participate were reduced by the restrictions and political debate over executive compensation of TARP recipients.

We first study the decision of banks to participate in CPP based on whether or not they submitted an application for a CPP equity injection. We find clear evidence of self-selection by banks. However, we find that strong banks rather than weak ones opted out of participating in CPP. Banks that opted out of CPP had strong capital ratios, had stable funding profiles, had high asset quality, and operated in better-performing regions than participating banks. Thus, the adverse signaling and debt overhang concerns raised by Hoshi and Kashyap (2010) do not appear to have prevented weak U.S. banks from participating in CPP infusions.

We then study the U.S. Treasury's approval decision for CPP applications. Our evidence suggests that capital injections under CPP occurred in a manner consistent with lowering financial distress costs. Banks that were approved for CPP injections were larger and posed greater systemic risk. However, they had significantly stronger asset quality than banks that were not approved for CPP injections, suggesting that capital was not provided to banks with high levels of troubled assets. Interestingly, we find that some banks declined CPP funds after being approved for an injection by the U.S. Treasury. These banks had substantially higher-quality assets and were located in better-performing regions than those that accepted capital injections. This suggests that better-performing banks viewed the capital injections to be relatively costly.

We examine whether banks rejected CPP injections to simply avail themselves of the certification effect of receiving U.S. Treasury approval. However, we find little evidence that CPP capital injections contained meaningful new information. Banks receiving CPP injections experienced significant valuation gains, but these gains occurred when CPP was initially announced and not when the banks were actually approved for the capital infusions. Strikingly, banks that announced that they had been approved for CPP infusions but rejected the funds also did not experience a meaningful change in equity values. Hence, adverse signaling problems appear not to have been an important concern that caused banks to refuse participation in the program. In contrast to the CPP approval process, the subsequent stress tests conducted under CAP entailed a much more detailed assessment of bank health. We find strong evidence of certification effects associated with these stress tests, indicating that the nature of the assessment conducted by regulators is important to investors in evaluating its information content.

Our results suggest that the debate and restrictions over CEO compensation played an important role in the capital infusion process. TARP contained limits on senior executive compensation, and the intense public and political controversy over compensation during this time also raised the prospect of broader government interference in compensation at banks. Highlighting the focus that regulators placed on executive compensation, we find that banks where CEOs earned more than \$500,000 annually were less likely to be approved for CPP injections. In addition, these banks were also more likely to reject TARP funds, suggesting that CEOs' personal costs from government restrictions were an important factor in whether banks accepted government support. We also study H.R. 1586, a bill that imposed a punitive 90% tax on all employee bonuses paid by institutions that had received \$5 billion or more in CPP capital. We show that stock prices of the affected banks dropped sharply around the passage of this bill. This result suggests that there were political interference costs for banks receiving TARP capital, possibly because the compensation restrictions may have given an advantage to non-TARP banks that could poach executives with unrestricted compensation packages.

We also examine the decision of banks to exit CPP. Banks that repaid CPP infusions tended to be larger, have stronger capital ratios, and have better asset quality than banks that continued to receive TARP support. In addition, we find that executive compensation is a strong determinant of whether a bank repays TARP funds. High levels of excess CEO compensation in the prior year substantially increase the likelihood that a bank will repay CPP. This suggests that the restrictions and scrutiny of executive compensation were an important determinant of banks' exiting decisions.

Our results suggest that CPP infusions may have had the unintended effect of prolonging banking recovery by changing investors' views about the government's incentives. Since CPP infusions were in the form of preferred stock, the amount of common equity in a bank's capital became the only

buffer protecting the value of the government's preferred equity claim. Thus, in assessing whether the government would seize a bank, investor focus shifted to the amount of the common equity in the bank's capital. As a result, even after the CPP infusions, bank valuations continued to decline as ongoing credit losses depleted the common equity capital of banks. We find that bank valuations rallied sharply around the assessment of adequacy of tangible common equity capital under CAP. These results highlight the importance that investors attach to the government's incentives in the design of a financial rescue program.

In related work, [Veronesi and Zingales \(2010\)](#) study the valuation effects of TARP for the initial CPP infusions that occurred on October 14, 2008. They find that the primary effect was to benefit bondholders and argue that other potential rescue strategies would have yielded larger net benefits. Unlike [Veronesi and Zingales \(2010\)](#), we study a comprehensive sample of banks that received CPP infusions that allows us to examine both the banks' participation and the Treasury's approval decisions along with their associated valuation effects. We also provide evidence on the subsequent stress tests under CAP and the decision of banks to exit CPP. However, we note that there are many aspects of the debate over optimal rescue packages that we are silent on. For example, [Hoshi and Kashyap \(2010\)](#) argue that capital injections need to be sufficiently large, but the limited variability in the capital infusion amounts under TARP does not permit us to explore this issue. We also do not address whether certain large banks with highly illiquid assets should have been dissolved or placed under government control, and we do not attempt to compare the cost or efficiency of the TARP capital injections to alternative strategies.

The article is organized as follows. Section 2 provides background on TARP. In Section 3, we examine CPP infusions and the incentives of banks to participate. In Section 4, we present the valuation effects of TARP capital infusions. Section 5 describes the changing political environment around TARP and its impact on CPP recipients. Section 6 discusses CPP repayments. Section 7 describes the changes in investor expectations about government incentives, and Section 8 concludes.

2. Background on the Troubled Asset Relief Program

TARP was conceived in response to the financial market turmoil that followed the collapse of Lehman Brothers and the government rescue of AIG in mid-September 2008. Table 1 provides a detailed timeline of the key events surrounding the capital infusions that occurred under TARP. TARP was enacted on October 3, 2008, and it authorized spending of \$700 billion to be implemented in two stages of \$350 billion each. As part of the first stage, the U.S. Treasury announced on October 13, 2008, that it would invest directly in equity of financial institutions. On October 14, 2008, CPP was announced, allocating

Table 1
Timeline of events related to TARP capital infusions

October 3, 2008: President Bush signs into law the Emergency Economic Stabilization Act of 2008, which establishes the \$700 billion Troubled Asset Relief Program (TARP).

October 13, 2008: U.S. Treasury interim assistant secretary Neel Kashkari announces a standardized program to purchase equity in a broad array of financial institutions.

October 14, 2008: The U.S. Treasury announces the Capital Purchase Program (CPP), which allows U.S. financial institutions to apply for a preferred stock investment by the U.S. Treasury. Nine large financial organizations announce they will subscribe to the facility in an aggregate amount of \$125 billion. The FDIC creates a new Temporary Liquidity Guarantee Program to guarantee the senior debt of all FDIC-insured institutions as well as deposits through June 30, 2009.

November 14, 2008: The deadline for publicly held financial institutions to apply for a CPP preferred stock investment by the U.S. Treasury.

February 4, 2009: The U.S. Treasury modifies restrictions on executive compensation for TARP participants. The total annual compensation for senior executive officers is limited to \$500,000, except for long-term restricted stock awards. Golden parachutes are prohibited for the top ten senior executives, and the next twenty-five executives are prohibited from receiving golden parachute payments greater than one year's compensation upon severance. Bonus claw-back provisions were extended from the top five executives to the next twenty-five executives.

February 10, 2009: U.S. Treasury Secretary Timothy Geithner announces the Financial Stability Plan involving forward-looking loss assessments for nineteen U.S. banks with assets exceeding \$100 billion, the Capital Assessment Program (CAP) whereby the U.S. Treasury will purchase preferred stock convertible to common equity in eligible banks, the creation of a Public-Private Investment Fund to acquire troubled loans and assets from financial institutions, expansion of the Federal Reserve's Term Asset-Backed Securities Loan Facility (TALF), and new initiatives to stem residential mortgage foreclosures and to support small business lending.

February 17, 2009: President Obama signs into law the American Recovery and Reinvestment Act of 2009, which includes a variety of spending measures and tax cuts intended to promote economic recovery. The act includes most of the executive compensation restrictions for TARP participants announced on February 4, 2009. The act includes additional compensation restrictions for TARP participants with a prohibition on bonuses, retention awards, or incentive compensation exclusive of long-term restricted stock awards that do not exceed one-third of the annual compensation.

February 23, 2009: The U.S. Treasury and federal bank regulatory agencies issue a joint statement in which they reiterate that under CAP, capital will be provided in the form of mandatorily convertible preferred stocks, and previous capital injections under CPP will also be eligible to be exchanged for these shares.

February 24, 2009: Federal Reserve chairman Ben Bernanke clarifies to Congress that the bank stress tests are not a precursor to nationalization of banks.

February 25, 2009: The U.S. Treasury publishes a white paper, term sheet, and FAQs containing details on the CAP program.

February 26, 2009: Iberiabank Corp. files notice with the Treasury Department that it will redeem all 90,000 outstanding shares of U.S. Treasury preferred stock for a redemption price of \$90.6 million.

March 19, 2009: The U.S. Congress proposes and passes H.R. 1586 in under twenty-four hours with an overwhelming margin. The bill imposes a punitive excise tax of 90% on all employee bonus payments made, retroactive to January 1, 2009, by institutions that have received \$5 billion or more in TARP capital.

March 31, 2009: Four bank holding companies (Marin Bancorp., Iberiabank Corp., Old National Bancorp., and Signature Bank) announce that they have redeemed all of the preferred shares issued to the U.S. Treasury under CPP.

April 24, 2009: The Federal Reserve Board publishes a white paper describing the process and methodology employed by federal banking supervisory authorities in their forward-looking assessment (stress test) of large U.S. bank holding companies.

May 6, 2009: The *Wall Street Journal* reports that results of the government's stress tests for several banks have been leaked to the media. It reports that American Express, Bank of New York Mellon, Capital One Financial, Goldman Sachs, MetLife, and JPMorgan Chase will not need to raise additional capital, while Bank of America needs \$34 billion, Wells Fargo requires \$15 billion, and GMAC needs \$11.5 billion. The *Journal* reports that Citigroup, State Street, Morgan Stanley, and Regions Financial will also need additional capital.

(continued)

Table 1
Continued

May 7, 2009: The Federal Reserve releases the results of the Supervisory Capital Assessment Program (SCAP) for the nineteen largest U.S. bank holding companies. The assessment finds that ten firms need to add \$185 billion to capital but that transactions and revenues since the end of 2008 have reduced the capital shortfall to \$75 billion. Banks that need to augment capital will be required to develop a detailed plan to be approved by its primary supervisor within thirty days and to raise the additional capital by November 2009.

June 10, 2009: The U.S. Treasury Department issues detailed rules regarding the limitations on executive and employee compensation imposed under the American Recovery and Reinvestment Act of 2009 for companies receiving funds under TARP. Treasury secretary Timothy Geithner also issues a press release on compensation principles.

December 31, 2009: The authority of the U.S. Treasury to purchase capital in financial institutions under CPP terminates, as specified in the Emergency Economic Stabilization Act of 2008.

\$250 billion for purchases of preferred stock of U.S. financial institutions.¹ On this day, nine financial institutions (henceforth referred to as the initial recipients)—Citigroup, Wells Fargo, JPMorgan, Bank of America, Goldman Sachs, Morgan Stanley, State Street, Bank of New York Mellon, and Merrill Lynch—received an aggregate infusion of \$125 billion.² These initial CPP recipients were chosen due to their status as the largest financial institutions and did not undergo a formal evaluation as part of the CPP process. Other banks had until November 14, 2008, to apply for CPP funds from the U.S. Treasury.

Under CPP, the U.S. Treasury would purchase nonvoting preferred stock, and banks could apply for this capital injection in amounts between 1% and 3% of their risk-weighted assets. CPP infusions took the form of preferred stock in order to be attractive to banks by virtue of being nondilutive to common shareholders. The U.S. Treasury would also receive warrants with a ten-year life to purchase common stock for an amount equal to 15% of the preferred equity infusion. The dividend on the preferred stock was set at 5%, but it would rise to 9% after three years. However, CPP infusions forbade dividend increases on common shares until the preferred shares were repaid.

CPP injections were accompanied by limits on executive compensation. The initial restrictions were outlined when CPP was introduced on October 14, 2008, and limited the tax deductibility of compensation for senior executives to \$500,000, required bonus claw-backs, and limited golden parachute payments. On February 4, 2009, the U.S. Treasury issued revised guidelines limiting total annual compensation for senior executives at TARP participants to \$500,000,

¹ Along with CPP, government guarantees on new bank debt issues and an increase in FDIC guarantee of non-interest-bearing transaction accounts were also unveiled on October 14, 2008. The bank debt guarantee was finalized on November 21, 2008, as the Temporary Liquidity Guarantee Program (TLGP).

² Of the nine institutions that received CPP infusions on October 14, 2008, three were investment banks until September 21, 2008, and did not historically report as bank holding companies. Hence, comparable financial statement data and capital ratios for these three institutions are unavailable, and we exclude them in analyses requiring financial characteristics. In addition to these nine institutions, Wachovia had signed a definitive merger agreement with Wells Fargo.

Table 2
Summary statistics on CPP capital infusions

Panel A: Allocation of Funds Under Capital Purchase Program (CPP) as of December 31, 2009

Type of Institution	Number of Banks	Amount (\$bn)
Commercial Banks	646	192.8
Public Banks	350	188.2
Private Banks	296	4.6
Thrifts	57	1.8
Non-bank Financial Institutions	4	10.3
Uncommitted Funds		45.1
Total	707	204.9

Panel B: Details on CPP Preferred Stock Injections

	Obs.	Mean	Median	25%	75%	Min	Max
CPP Infusion, \$mill	350	537.8	22.0	8.2	72.3	1.0	25,000
CPP Infusion / RWA, %	341	2.9	3.0	2.9	3.0	1.0	10.5
CPP Infusion / MVE, %	339	29.7	26.6	17.5	38.4	5.0	115.0
Pre-CPP Tier 1 ratio, %	341	10.9	10.2	9.4	11.7	6.3	33.6
Post-CPP Tier 1 ratio, %	341	13.8	13.1	12.3	14.6	8.9	36.6

Panel A shows the allocation of CPP funds to banks until the closing of the program on December 31, 2009. Panel B presents summary statistics on the amount of CPP preferred equity capital injections in public commercial banks. Bank financial data are as of September 30, 2008. *RWA* is risk-weighted assets. *MVE* is market value of equity as of September 30, 2008. *Pre-CPP Tier 1 ratio* is Tier 1 ratio on September 30, 2008. *Post-CPP Tier 1 ratio* is Tier 1 ratio on September 30, 2008, plus the TARP infusion scaled by risk-weighted assets.

excluding certain incentive awards along with additional limits on severance payments. These guidelines were further modified on February 17, 2009, under the American Recovery and Reinvestment Act (ARRA), which prohibited bonuses, retention awards, and incentive compensation other than long-term restricted stock awards that did not exceed one-third of annual compensation.

Panel A of Table 2 shows that CPP capital was provided to 707 financial institutions, including 646 commercial banks. Of these, 350 are publicly traded banks and account for \$188 billion of the \$193 billion spent on commercial banks. Panel B of Table 2 shows that the average infusion of preferred stock is \$538 million, but the median size is much lower, at \$22 million. Almost all of the variation in infusion amounts is dictated by the terms of CPP, which allowed banks to obtain between 1% and 3% of risk-weighted assets (RWA). Most banks applied for the maximum amount, as illustrated by the ratio of CPP infusion, to RWA of 2.9% for the twenty-fifth percentile of this distribution.³ Regulatory capital ratios for CPP recipients improved materially after the infusions, with the median Tier 1 ratio rising from 10.9% to 13.8%.

Although Tier 1 ratios improved considerably with the CPP injections, the common equity component of banks' capital was unaffected. In the ensuing months, the equity capital of banks continued to be depleted as banks faced

³ In a handful of cases, banks received more than 3% of RWA, usually in the context of a government-supported acquisition. For example, PNC Financial Services received 6.5% of RWA because its infusion occurred to support its purchase of Hampton Roads Bancshares, which in turn received 10.5% of RWA.

ongoing credit losses and write-downs. As a result, stock prices of banks continued to fall sharply through the first quarter of 2009, prompting further government intervention in the banking sector. The second stage of TARP sought to address this ongoing banking crisis. On February 10, 2009, the government introduced the Financial Stability Plan, funded by the remaining \$350 billion allocation under TARP. This plan had four components, including CAP for banks, a consumer lending initiative, a public-private investment fund for illiquid assets, and a home mortgage modification initiative.

Under CAP, nineteen banks with assets greater than \$100 billion were subject to a detailed, forward-looking assessment of capital needs called the Supervisory Capital Assessment Program (SCAP). Banks deemed to be deficient in capital under SCAP would be provided access to a new funding mechanism from the U.S. Treasury that would inject common equity and serve as a bridge to raising capital from public markets. On February 23–25, 2009, important details about the goals of CAP were disclosed, and the nature of the assessment, which included a stress test under an adverse macroeconomic scenario, was described. On April 24, the SCAP methodology was disclosed in substantial detail. The SCAP results for several banks were leaked on May 6, and the results were officially released on May 7. Ten of the nineteen banks subject to SCAP were required to raise an aggregate amount of \$75 billion in new equity.

There were several notable differences between the CPP and SCAP processes. First, with the exception of the initial nine recipients, participation in CPP was voluntary, whereas SCAP was mandatory for the affected banks. Thus, under CPP, banks could self-select to receive government capital infusions by evaluating potential benefits and costs. Second, the SCAP process was more transparent, with disclosure about which banks were subject to the evaluation and substantial detail about the methodology. Third, capital infusions under SCAP were in common equity (in the form of convertible shares) instead of preferred stock. As we discuss later, government ownership of preferred equity may increase investors' perceived risk of future regulatory intervention in banks and impede recovery of the sector. Fourth, there was a detailed evaluation of each bank's financial health under SCAP; banks had to submit details of their assets and loans and their expected performance that had previously not been disclosed to regulators. Finally, the government's role in SCAP was largely supervisory in nature, and its role as a capital provider was to serve as a bridge to nongovernment sources. Thus, the certification effect of the SCAP process is likely to be greater than that for CPP infusions.

Since the introduction of TARP, there has been substantial public and political debate about the capital infusions. Political sentiment against the infusions heightened after revelations of excessive spending and perquisite consumption at some TARP recipients, including an expensive office refurbishment at Merrill Lynch, the planned purchase of a corporate jet by Citigroup, and a lavish corporate retreat by Northern Trust. Sparked by media outcry

over retention bonuses paid by AIG, on March 19, 2009, the U.S. Congress proposed and passed H.R. 1586, a controversial bill that retroactively imposed a punitive excise tax of 90% on all employee bonus payments by institutions that had received \$5 billion or more in TARP funds. Citing concerns over the restrictions imposed by TARP, on March 31, 2009, four banks announced that they had repaid all of the preferred shares issued to the U.S. Treasury under CPP, and in the following months, many other banks submitted applications to repay CPP infusions. New infusions of equity ended on December 31, 2009, when the authority of the U.S. Treasury to purchase shares in financial institutions under CPP expired.

3. How Was CPP Capital Allocated?

To examine the determinants of CPP capital infusions, we begin with a sample of 590 publicly traded banks that have annual and quarterly consolidated financial statements along with data on executive compensation from SNL Financial. Due to concerns that investors may interpret the non-award of CPP capital as a negative signal that may trigger a bank run, applicants to the CPP program have not been publicly disclosed by the U.S. Treasury. Therefore, we conduct a detailed search of publicly filed financial statements, press releases, news wires, and company websites for any disclosure by banks about their CPP applications.

Of the 590 banks in the sample, we find 121 banks that disclose that they are not applying for funds under CPP. We refer to them as nonparticipating banks and the others as participating banks. Of the 469 participating banks, 268 banks received CPP infusions based on data from the U.S. Treasury. However, we also find 51 banks that disclose that they received approval for CPP infusions but opted not to receive the funds. Therefore, our sample contains 319 banks that received approval for infusions from the U.S. Treasury. We also find 49 banks that disclosed that they had applied for CPP but do not appear on the U.S. Treasury list of banks approved for CPP infusions. We treat these 49 banks along with the remaining 101 banks for whom CPP status is unknown as banks that did not receive approval from the U.S. Treasury.

The data indicate a sequential decision-making process in which banks decide whether to participate in the application process, the U.S. Treasury approves or disapproves the application, and banks decide whether to accept or reject the funds. Our empirical strategy for analyzing the CPP infusion process follows this pattern. We use a sequential logit model to first estimate a bank's decision to apply for the program, then estimate the Treasury's approval decision, and finally estimate the bank's decision to accept or reject CPP funds.⁴

⁴ The sequential logit model assumes that the decision process consists of sequential and independent stages of choices. The independence assumption requires that the utility from subsequent decisions does not influence the initial choice, which appears reasonable in our setting given the lack of transparency about the CPP

3.1 Bank characteristics and control variables

We use several bank characteristics in the tests on the CPP infusion process, and we describe the construction of these variables below. We consider two measures of the systemic risk that banks posed. The first variable is bank size, measured by total bank assets. We also use the notional value of outstanding derivative contracts as a measure of the potential magnitude of counterparty risk that a bank may impose on the financial system.

We consider two measures of capital adequacy—the Tier 1 ratio and the tangible common equity ratio. The Tier 1 ratio is the most common regulatory capital metric and was bolstered by CPP infusions. The tangible common equity ratio removes the effects of hybrids, preferred stock, quasi-debt instruments, and intangible assets from the Tier 1 ratio and considers only the components of Tier 1 capital that represent tangible common equity. Since CPP infusions were in the form of preferred stock, tangible common equity ratios remained largely unchanged as a result of the injections.

We consider the funding mix employed by banks. [Ivashina and Scharfstein \(2010\)](#) note that deposits were regarded as a more stable source of funding that lowered banks' financial exposure during the financial crisis. Hence, we calculate the wholesale debt ratio, which is the ratio of nondeposit liabilities to assets and represents the proportion of nondeposit borrowing by a bank.

To evaluate asset quality, we calculate the net charge-offs ratio (which measures the realized losses on loans) and the loan loss provisions ratio (which measures the amount reserved to cover near-term expected loan losses). While net charge-offs measure the current losses due to defaults, loan loss provisions provide a forward-looking measure of expected loan losses. In addition to loans, which make up the vast majority of assets for a typical bank, a few banks also hold trading assets that are marked to market. To assess the quality of these assets, we calculate the magnitude of write-downs in the investment portfolio from the third quarter of 2007 through the third quarter of 2008 for each bank.

We consider two measures of executive compensation. First, we calculate the log of the excess of a CEO's total compensation over \$500,000 because banks accepting TARP funds faced higher costs from the nondeductibility for tax purposes of compensation exceeding this level, according to the initial CPP rules and from the \$500,000 limit announced subsequently. Since the limits on compensation likely prompted broader concern among CEOs about future restrictions that might be imposed, we also construct a measure of excess CEO compensation. Specifically, we define excess CEO compensation as the residual from a regression of the natural logarithm of the total CEO compensation on the natural logarithm of bank assets, the return on assets

approval process. A potential alternative approach is the nested logit model, which assumes that the decision process considers the entire choice set simultaneously. However, the nested logit model assumes that errors are uncorrelated between groups of choices. This assumption is likely to be violated in our setting, where banks decline participation in CPP in the first stage and also decline to accept CPP funds in the final stage.

in that year, and the bank's age. We posit that CEOs with high excess compensation are likely to be concerned about government interference in compensation, making them less likely to participate in CPP and more likely to decline funds following Treasury approval. We define total CEO compensation as the sum of salary, bonus, value of stock, and option awards granted during the year and other non-equity incentive plan compensation. All compensation data are as of the end of 2008.

Duchin and Sosyura (2010) argue that the political influence of banks was an important determinant of how CPP funds were allocated. Following Duchin and Sosyura, we examine whether a bank's executive is a member of the Board of the Federal Reserve in either 2008 or 2009. Each of the twelve Federal Reserve Banks has its main bank and up to four branches. Executives of the member banks in a given district represent three out of nine directors on the boards of each main bank and one or two directors on the boards of branches. In our sample, forty-three banks had their representatives on the boards of the Federal Reserve Banks. We construct an indicator variable, Fed Board, that equals one if a bank has a representative on the Federal Reserve boards. The second measure of political influence is based on political contributions. For each bank we calculate the total amount of contributions made by bank-sponsored political action committees and bank employees to members of the House Financial Services Committee in the 2008 election campaign and scale it by the bank's assets. These data are collected from the Center for Responsive Politics.

We account for geographic variation in bank activities that may have influenced CPP infusions since the impact of the U.S. recession was not uniform across the country. Banks in severely affected regions likely had stronger incentives to apply for CPP funds, and regulators may have given these banks preferential treatment. Since many banks operate in multiple states, we control for the change in the economic performance of each state that a bank operates in. For each state, we calculate the change in its Federal Reserve macroeconomic index from the start of the recession in December 2007 through October 2008. This macroeconomic index tracks nonfarm payroll employment, the average hours worked in manufacturing, the unemployment rate, and wage and salary disbursements at the state level. We weight the percentage change in the state-level macroeconomic index by the share of deposits a bank holds in each state to arrive at a deposit-weighted measure of the change in regional economic conditions faced by each bank and include this variable in our tests. In addition, the models are estimated with state-fixed effects for the state in which the bank is headquartered.

We control for each bank's loan composition. Regulators may have incentives to support banks that specialize in certain types of lending if continued supply of credit to that activity was considered to be important. For example, it is possible that banks specializing in real estate lending were more likely to

be supported in order to prevent a reduction in credit that might exacerbate housing price declines. We define real estate loans as all commercial and residential real estate loans, commercial and industrial (C&I) loans as loans to corporations and businesses, and consumer loans as credit card loans, loans under revolving credit plans, auto and student loans, and other installment or single-payment consumer loans.

3.2 CPP capital injections: Bank participation decision

Columns (1) and (2) of Table 3 examine a bank's decision to participate in CPP using a logit model in which the dependent variable is set to one for the 469 participating banks and zero for the 121 nonparticipating banks in the sample. Our reliance on voluntary disclosures by banks about their application decision creates the likelihood that bank participation is observed with noise. To the extent that we measure participation imperfectly, the precision of the coefficient estimates is lowered.

We find that the probability of bank participation is negatively related to capital adequacy, with the coefficient on the Tier 1 ratio negative and significant at the 1% level. The point estimates imply that a one-standard-deviation increase in the Tier 1 ratio lowers the probability of participation by 8.1 percentage points. Since the unconditional likelihood of participation in the sample is 80%, a one-standard-deviation increase in Tier 1 lowers participation likelihood by about one-tenth of the unconditional likelihood. Banks with more vulnerable funding strategies are more likely to participate, as indicated by the positive coefficient on wholesale funding. A one-standard-deviation increase in the wholesale debt ratio increases participation probability by 4.3 percentage points. Finally, the macroeconomic environment in the states where a bank operates has a large impact on participation likelihood. A one-standard-deviation increase in the state macro index growth is associated with a participation probability that is 23.7 percentage points lower. Overall, these results suggest that banks with low expected distress costs, as indicated by high capital levels, more stable deposit-based funding, and a better operating environment, are less likely to participate in CPP.

3.3 CPP capital injections: Treasury approval decision

Columns (3) and (4) of Table 3 report logit model estimates for the Treasury approval decision where the dependent variable equals one if the bank received approval for a CPP capital injection and is zero otherwise. Banks that did not apply to CPP are excluded from the analysis in these specifications.

Banks with higher potential for systemic risk were more likely to be approved for CPP injections. We find that bank size has a positive and significant effect on the probability of being approved. In addition, the size of a bank's derivatives portfolio is also positively related to the probability of

Table 3
Bank participation in CPP and Treasury approval decision

	Bank Participation Decision (All banks)		Treasury Approval Decision (Participating banks)	
	(1)	(2)	(3)	(4)
Ln(Total Assets)	-0.12 (0.43)	-0.13 (0.39)	0.82*** (0.00)	0.84*** (0.00)
Derivatives/Assets, %	0.05 (0.19)	0.05 (0.19)	0.09** (0.05)	0.08* (0.07)
Wholesale Debt, %	0.05** (0.03)	0.04** (0.03)	0.04* (0.08)	0.05** (0.05)
Tier 1 Ratio, %	-0.11*** (0.00)	-0.12*** (0.00)	-0.03 (0.47)	-0.01 (0.75)
Net Charge-offs, %	0.20 (0.40)		-1.04*** (0.00)	
Loan Loss Provisions, %		0.24 (0.19)		-0.82*** (0.00)
Write-downs, %	2.51 (0.23)	2.62 (0.23)	-8.16* (0.08)	-7.48* (0.09)
Fed Board	0.54 (0.48)	0.55 (0.46)	0.54 (0.56)	0.50 (0.57)
Campaign Contributions, %	1.14 (0.29)	1.11 (0.31)	0.46 (0.64)	0.60 (0.53)
CEO Compensation > 500K	0.01 (0.71)	0.01 (0.75)	-0.08** (0.04)	-0.08* (0.06)
Excess CEO Compensation	-0.32 (0.18)	-0.31 (0.20)	0.28 (0.33)	0.24 (0.40)
Real Estate / Total Loans, %	0.05 (0.13)	0.04 (0.18)	-0.01 (0.79)	-0.00 (0.94)
C&I / Total Loans, %	0.07* (0.05)	0.07* (0.08)	0.05 (0.26)	0.07 (0.18)
Consumer / Total Loans, %	0.03 (0.44)	0.03 (0.50)	0.03 (0.53)	0.03 (0.54)
State Macro Index Growth, %	-0.72* (0.10)	-0.74* (0.09)	-0.30 (0.58)	-0.28 (0.57)
HQ State Fixed Effects	Yes	Yes	Yes	Yes
Observations	535	535	423	423
Pseudo R-squared	0.18	0.18	0.26	0.27

Columns (1) and (2) present logit model estimates for the likelihood that a bank submits an application for CPP capital injections. The dependent variable equals one for participating banks and zero for banks that opted out of applying for CPP funds. Columns (3) and (4) present logit model estimates for the likelihood that a bank's application for CPP injections is approved by the U.S. Treasury for the sample of participating banks. The dependent variable equals one for the banks that were approved by the Treasury and zero for the banks that applied but did not get approval. Financial data are as of September 30, 2008. *Ln(Total Assets)* is the natural logarithm of assets. *Derivatives / Assets, %* is the notional value of the bank's derivatives contracts scaled by assets. *Wholesale Debt, %* is the sum of subordinated notes and debentures and other borrowed money scaled by assets. *Tier 1 Ratio, %* is Tier 1 capital scaled by risk-weighted assets. *Write-downs, %* are realized and mark-to-market losses on trading assets from September 30, 2007, to September 30, 2008, scaled by assets. *Net Charge-offs, %* is credit losses from September 30, 2007, to September 30, 2008, net of recoveries scaled by the average total loans. *Loan Loss Provisions, %* is the amount reserved from September 30, 2007, to September 30, 2008, scaled by average loans. *Campaign Contributions, %* is political contributions made by bank-sponsored political action committees and employees to the members of the House Financial Services Committee in the 2008 election campaign. *Fed Board* is an indicator for the financial institutions that had executives on the boards of the Federal Reserve Banks in 2008 and 2009. *CEO Compensation > 500K* is the natural logarithm of the difference between the total CEO compensation and \$500,000 if it is positive and zero otherwise. *Excess CEO Compensation* is the residual from the regression of the logarithm of total CEO compensation in 2008 on the logarithm of the bank's total assets, return on assets, and age. *C&I / Total Loans, %* is the fraction of commercial and industrial loans in total loans. *State Macro Index Growth, %* is calculated as the weighted average of the changes in the Federal Reserve Bank of Philadelphia's state-coincident indexes from December 2007 to October 2008 with the share of deposits of a given bank in a given state used as weights. *HQ State Fixed Effect* is an indicator equal to one for the state in which the bank is headquartered and zero otherwise. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. *p*-values based on heteroscedasticity-adjusted standard errors are reported in parentheses.

being approved for CPP funds. A one-standard-deviation increase in the log of bank assets increases the probability of Treasury approval by three percentage points, and a corresponding change in derivatives holdings increases it by 32 percentage points. The effect of derivatives is economically large compared with the unconditional probability of Treasury approval of 68% in the sample.

Banks facing greater exposure to financial markets through their funding strategies were more likely to be approved; the coefficient on the wholesale debt ratio is positive in both models. A one-standard-deviation increase in the wholesale debt ratio is associated with a 5.9-percentage-point higher probability of approval. However, we do not find a significant effect of Tier 1 ratio, suggesting that CPP funds were not directed toward banks with weak capital ratios. In untabulated tests, we find a similar result if we use the tangible common equity ratio instead.

CPP recipients have better-performing assets than nonrecipient banks. Model (3) shows that the coefficient on the net charge-offs ratio is negative and significant at the 1% level, indicating that high credit losses lower the probability of receiving CPP approval. A similar result is obtained in model (4), where the coefficient on the loan loss provisions ratio is negative and significant at the 1% level, suggesting that expectation of high future loan losses also lowers the probability of CPP approval. We also find that asset write-downs are negatively related to CPP approval. The economic magnitude of these effects is large. A one-standard-deviation increase in charge-offs (provisions) lowers approval probability by 17.9 percentage points (20.7 percentage points), while a corresponding increase in write-downs lowers it by 24 percentage points. Thus, asset quality was an important determinant of the approval process, and funds were provided to banks with better-performing assets rather than to institutions with troubled assets. This suggests that TARP attempted to support economically healthy banks rather than bailing out banks with distressed assets.

CEO compensation in excess of \$500,000 lowers the likelihood of CPP approval. Thus, executive compensation appears to have played a role in the allocation process beyond the tax deductibility limits contained in the initial CPP rules. The economic magnitude of the effect is sizeable. A one-standard-deviation increase in the log of CEO compensation in excess of \$500,000 is associated with an 11.4-percentage-point reduction in Treasury approval, or roughly one-sixth of the size of the unconditional approval probability.

A bank's exposure to state-level macroeconomic conditions has no effect on CPP approval likelihood, suggesting that geographic variation in economic conditions was not an important determinant of capital infusions. In contrast to [Duchin and Sosyura \(2010\)](#), we also find no evidence that measures of political influence affect Treasury approval decisions.

We examine whether the results are driven by the very large banks that were the initial recipients of CPP funds on October 14, 2008. In untabulated results,

we repeat the analysis, excluding the commercial banks in our sample that were among the initial recipients, and obtain virtually identical conclusions.

3.4 CPP capital injections: Bank rejection decision

Hoshi and Kashyap (2010) argue that a financial rescue plan should encourage broad participation by lowering banks' incentives to reject government support. Banks can have incentives to reject support because infusions may send an adverse signal. In addition, a debt overhang problem lowers equity holder incentives to receive support since the gains in the event of a recovery may accrue largely to debt holders. Finally, the prospect of government interference in executive compensation provided banks with another incentive to reject TARP funds. The enhanced restrictions on compensation introduced in February 2009, along with the ensuing public and political debate on compensation, may have led executives to reassess their personal costs of participating in TARP. Since banks were allowed to reject infusions after receiving Treasury approval, it is also possible that some banks applied for and rejected CPP funds as a means to avail themselves of certification associated with Treasury approval.

As a first step in understanding these explanations, we estimate the likelihood that a bank rejects CPP injections after having received Treasury approval. Table 4 presents logit estimates of the bank rejection decision. The dependent variable equals one if a bank rejects CPP capital and zero if a bank accepts CPP funds. These models are estimated on the subsample of firms that received Treasury approval for CPP infusions.

All specifications show that large banks were less likely to reject CPP injections. Untabulated results show that the effect of bank size persists if we exclude the six initial commercial bank recipients. However, the economic importance of size appears to be modest. A one-standard-deviation increase in size is associated with a 0.4-percentage-point change in rejection probability, compared with the unconditional probability of rejection in the sample of 16%.

Banks with weak asset quality are also less likely to reject CPP injections. The coefficients on the net charge-offs and loan loss provisions ratios are negative and significant in all specifications. A one-standard-deviation increase in net charge-offs is associated with a 7.5-percentage-point decrease in rejection probability, or almost one-half the size of the unconditional rejection probability. A corresponding increase in loan loss provisions is associated with a 6.5-percentage-point decrease in rejection probability. Thus, the quality of loans held by a bank is an important determinant of its decision to accept or reject capital infusions.

Compensation-related incentives appear to influence the decision of banks to accept CPP funds. Models (1) and (2) show that banks with CEO compensation over \$500,000 were likely to reject CPP infusions. Models (3) and (4) show that excess CEO compensation is also positively related to the likelihood of

Table 4
Bank decision to reject CPP funds

	Bank Rejection Decision (Banks approved for CPP infusions)			
	(1)	(2)	(3)	(4)
Ln(Total Assets)	-0.97*** (0.00)	-0.96*** (0.00)	-0.64*** (0.00)	-0.65*** (0.00)
Derivatives/Assets, %	0.01* (0.06)	0.01** (0.03)	0.01 (0.11)	0.01** (0.05)
Wholesale Debt, %	-0.01 (0.75)	-0.01 (0.78)	-0.01 (0.75)	-0.01 (0.76)
Tier 1 Ratio, %	0.01 (0.82)	0.05 (0.28)	0.01 (0.76)	0.05 (0.23)
Net Charge-offs, %	-1.65* (0.06)		-1.71** (0.04)	
Loan Loss Provisions, %		-1.51*** (0.01)		-1.60*** (0.00)
Write-downs, %	-1.94 (0.60)	-2.24 (0.55)	-2.00 (0.59)	-2.35 (0.52)
Campaign Contributions, %	-1.33 (0.51)	-0.99 (0.62)	-1.95 (0.33)	-1.58 (0.43)
CEO Compensation > 500K	0.11** (0.03)	0.11** (0.04)		
Excess CEO Compensation			1.15** (0.03)	1.18** (0.02)
Real Estate / Total Loans, %	0.04 (0.38)	0.05 (0.27)	0.05 (0.31)	0.06 (0.21)
C&I / Total Loans, %	0.03 (0.60)	0.04 (0.43)	0.04 (0.44)	0.06 (0.30)
Consumer / Total Loans, %	0.04 (0.50)	0.05 (0.44)	0.07 (0.32)	0.08 (0.28)
State Macro Index Growth, %	1.29** (0.01)	1.28** (0.02)	1.17** (0.04)	1.18** (0.04)
HQ State Fixed Effects	Yes	Yes	Yes	Yes
Observations	209	209	209	209
Pseudo R-squared	0.22	0.23	0.22	0.23

This table presents logit model estimates for the likelihood of rejection of CPP funds for banks that received approval from the Treasury. The dependent variable is equal to one for banks that received Treasury approval but rejected CPP funds and is zero for other CPP recipients. Financial data are as of September 30, 2008. *Ln(Total Assets)* is the natural logarithm of assets. *Derivatives / Assets, %* is the notional value of the bank's derivatives contracts scaled by assets. *Wholesale Debt, %* is the sum of subordinated notes and debentures and other borrowed money scaled by assets. *Tier 1 Ratio, %* is Tier 1 capital scaled by risk-weighted assets. *Write-downs, %* are realized and mark-to-market losses on trading assets from September 30, 2007, to September 30, 2008, scaled by assets. *Net Charge-offs, %* is credit losses from September 30, 2007, to September 30, 2008, net of recoveries scaled by the average total loans. *Loan Loss Provisions, %* is the amount reserved from September 30, 2007, to September 30, 2008, scaled by average loans. *Campaign Contributions, %* is political contributions made by bank-sponsored political action committees and employees to the members of the House Financial Services Committee in the 2008 election campaign. *Fed Board* is an indicator for the financial institutions that had executives on the boards of the Federal Reserve Banks in 2008 and 2009. *CEO Compensation > 500K* is the natural logarithm of the difference between the total CEO compensation and \$500,000 if it is positive and zero otherwise. *Excess CEO Compensation* is the residual from the regression of the logarithm of total CEO compensation in 2008 on the logarithm of the bank's total assets, return on assets, and age. *C&I / Total Loans, %* is the fraction of commercial and industrial loans in total loans. *State Macro Index Growth, %* is calculated as the weighted average of the changes in the Federal Reserve Bank of Philadelphia's state-coincident indexes from December 2007 to October 2008, with the share of deposits of a given bank in a given state used as weights. *HQ State Fixed Effect* is an indicator equal to one for the state in which the bank is headquartered and zero otherwise. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. *p*-values based on heteroscedasticity-adjusted standard errors are reported in parentheses.

CPP rejection. Thus, the compensation limits imposed by TARP as well as the specter for further political interference appear to have influenced banks' decisions to accept capital injections.⁵ However, the economic importance of compensation variables is relatively modest, with a one-standard-deviation increase in both variables leading to a 0.4-percentage-point increase in rejection likelihood, or 2.5% of the unconditional rejection probability.

Economic activity in the geographical footprint of the banks is an important determinant of banks' rejection decisions. Banks are more likely to reject government support when the state macroeconomic index increases more. A one-standard-deviation increase in the state macroeconomic index growth leads to a 36.6-percentage-point increase in rejection likelihood.

Overall, the evidence suggests that CPP capital was allocated in a manner consistent with the stated objectives of the program. Banks that were more systemically important, as measured by their size and derivatives exposures, were more likely to receive approval for capital infusions, as were banks with funding profiles that made them exposed to variability in financial market conditions. Banks with strong regulatory capital ratios opted out of the program, and among the banks that participated, injections do not appear to have been targeted toward banks with weak capital ratios. Banks with strong asset quality opted out of applying for infusions, and CPP injections flowed to banks with strong asset quality, suggesting that the program did not attempt to prop up banks with troubled assets. We also find that banks perceived CPP injections to have associated costs. The restrictions over executive compensation and broader concerns over government influence in compensation decisions appear to have deterred CEOs whose compensation packages made them vulnerable to restrictions from participating in the program.

4. Certification Effects of TARP Capital Infusions

A possible goal of a financial rescue plan is to certify to investors that an institution is healthy, easing informational asymmetries that would constrain raising new capital from outside investors. Consistent with this goal, Henry Paulson suggested that "such a program would be designed to encourage the raising of new private capital to complement public capital."⁶ If CPP infusions were designed to certify the health of the bank, the announcement of the equity infusion should convey positive news. However, if TARP was used to support weak institutions, CPP infusions may convey negative information about a bank's financial health and may discourage participation by banks.

⁵ Alternatively, it is possible that if CEO compensation is determined optimally (John, Mehran, and Qian 2010), excess compensation reflects the effect of unobserved CEO ability. Under this interpretation, high-ability CEOs rejected TARP support perhaps because they were confident of their ability to withstand the financial crisis. However, this reasoning suggests that CEO ability should be more important in explaining a bank's decision to apply to TARP instead of explaining their rejections following Treasury approval.

⁶ Paulson says the U.S. Treasury will buy bank equity "as soon as we can"; *Bloomberg News*, October 10, 2008.

Some evidence on the valuation effects of TARP infusions is provided by [Veronesi and Zingales \(2010\)](#), who study the initial recipients of CPP capital on October 14, 2008.⁷ However, certification effects cannot be relevant for the initial recipients, since these banks were chosen because of their size and did not undergo a formal screening process by the U.S. Treasury.

We examine the valuation effects of capital infusions under TARP by studying both CPP infusions as well as the capital adequacy assessments under SCAP using standard event study methodology.⁸ The lack of trading prices for bonds and active CDS contracts for most banks precludes us from calculating changes in the value of debt. Hence, unlike [Veronesi and Zingales \(2010\)](#), we are unable to assess whether the injections created more value than they cost taxpayers.

4.1 Valuation effects of CPP announcements

Panel A of Table 5 reports raw and excess stock returns associated with the CPP infusions. Excess returns are calculated using market model parameters estimated against the S&P 500 index from January 1, 2007, to September 17, 2008, the date before plans to support the financial system were revealed. We

Table 5
Capital Purchase Program (CPP) and stress-test (SCAP) announcements

Panel A: CPP

	Initial Recipients		Later Recipients	
	Raw Return,%	CAR,%	Raw Return,%	CAR,%
October 14, 2008: CPP Announcement				
Mean	13.94***	14.94***	3.48***	3.96**
Median	(16.41***)	(17.44***)	(0.75***)	(1.29***)
Obs.	9	9	223	223
CPP Approvals				
Mean			0.13	0.46
Median			(0.00)	(0.20)
Obs.			223	223
CPP Nonparticipations and Rejections				
	Banks not participating in CPP		Banks rejecting CPP	
	Raw Return,%	CAR,%	Raw Return,%	CAR,%
Mean	0.18	0.59	0.63	0.64
Median	(-0.98)	(-0.12)	(-0.10)	(0.01)
Obs.	32	32	19	19

(continued)

⁷ The initial returns around October 14, 2008, are potentially contaminated by other initiatives that were announced concurrently, such as the bank debt guarantee program and the extension of FDIC insurance, though [Veronesi and Zingales \(2010\)](#) estimate the value of these guarantees to be relatively modest.

⁸ A potential concern is that the capital infusions might be partially predictable given our earlier evidence on the determinants of the CPP approval process. To assess this, we follow the approach in [Acharya \(1998\)](#) and use our logit model estimates in Table 3 to calculate the conditional probability of approval. However, in most specifications, the conditional probability is not significantly related to the observed excess returns, implying that the conditional event study approach is not suitable in this setting.

Table 5
Continued
Panel B: SCAP

	Banks Subject to Stress Test		Banks Exempt from Stress Test		Banks with Capital Shortfall		Banks w/o Capital Shortfall	
	Raw Return,%	CAR,%	Raw Return,%	CAR,%	Raw Return,%	CAR,%	Raw Return,%	CAR,%
February 10, 2009: Announcement of the Financial Stability Plan								
Mean	−15.60***	−7.26***	−4.06***,A	−1.03***,A				
Median	(−13.33***)	(−5.32***)	(−4.16***),A	(−0.49***),A				
Obs.	18	18	398	398				
February 23–24, 2009: Clarification About Capital Assistance Program								
Mean	17.43***	17.10***	0.89*,A	0.78 ^A	24.04***	23.66***	10.82***,A	10.55***,A
Median	(16.22***)	(16.43***)	(0.18*),A	(0.30*),A	(22.69***)	(21.95***)	(9.92***),A	(9.95***),A
Obs.	18	18	349	349	9	9	9	9
April 24, 2009: Details of Stress Test (SCAP) Methodology								
Mean	4.21**	1.38	3.35***	2.33**	3.20	0.31	5.21*	2.45
Median	(3.22**)	(0.54)	(1.92***)	(0.31***)	(3.17)	(0.09)	(3.27**)	(0.98)
Obs.	18	18	384	384	9	9	9	9
May 6–7, 2009: Results of Stress Test (SCAP)								
Mean	7.71***	7.32***	1.36***,B	1.19***,B	8.36**	8.13**	7.06	6.50
Median	(5.64***)	(5.23***)	(0.25***),A	(0.18*),A	(6.40**)	(5.97**)	(2.97*)	(2.61)
Obs.	18	18	333	333	9	9	9	9

This table presents raw and abnormal returns on the days of announcements related to CPP and SCAP. Panel A reports announcement returns for initial and later CPP recipients on the day of announcement of CPP (October 14, 2008) and on the days of announcement of U.S. Treasury approvals for CPP recipients. *Initial Recipients* are the nine banks that were provided CPP injections on October 14, 2008. *Later Recipients* are banks for which capital infusions were approved between October 21, 2008, and December 31, 2009. Panel A also reports returns on the days of announcement of the decision not to apply for CPP funds and on the days of announcement of the decision not to accept CPP funds after being approved by the U.S. Treasury. Panel B reports returns on the days of four announcements related to SCAP: announcement of the Financial Stability Plan (February 10, 2009); clarification (February 23–24, 2009); details of the stress-test methodology (April 24, 2009); and the stress-test results (May 6 and May 7, 2009). CAR is the abnormal return calculated using return on S&P 500 as the market index and market model with parameters estimated over the period from January 1, 2007, until September 17, 2008. *Banks Subject to Stress Test* are U.S. banks with year-end 2008 assets exceeding \$100 billion that were required to participate in the Supervisory Capital Assessment Program (SCAP). *Banks Exempt from Stress Test* are all other public banks that have nonzero trading volume on event dates. *Banks with Capital Shortfall* is a subsample of the banks subject to stress test that were required to raise additional capital. *Banks w/o Capital Shortfall* is a subsample of the banks subject to stress test that were not required to raise additional capital. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. ^A, ^B, and ^C indicate statistical significance of the differences between subsamples at the 1%, 5%, and 10% levels, respectively.

require nonzero trading volume on a particular day for a bank to be included in our analysis of the announcement.

We examine common stock returns around October 14, 2008, as well as on the announcement date of CPP capital infusions for later recipients. Excess returns are strongly positive for the initial nine recipients on October 14, 2008, as shown in Panel A of Table 5. The average $(-1,0)$ excess return is 14.9%, with a median of 17.4%, and both are statistically significant at the 1% level.⁹

Panel A of Table 5 shows that later CPP recipients also experienced significant positive excess returns on October 14, 2008, even though the fact that these institutions would receive CPP capital was not known at the time. One explanation is that the market anticipated that these banks would receive capital infusions. An alternative explanation is that the CPP program improved equity valuations of all banks by lowering systemic risk in the financial system. Under either interpretation, the positive excess returns on October 14, 2008, for the later recipients do not support a certification effect.

Excess returns around the announcement of capital infusions for the later recipients are statistically insignificant. Thus, CPP capital infusions do not convey a certification effect, but also do not appear to be accompanied by an adverse signaling effect that may discourage bank participation.¹⁰ The non-negative announcement returns for CPP injections differ from the evidence on capital raised by banks from nongovernment sources. Cornett and Tehranian (1994) document an average CAR $(-1,0)$ of -1.56% around announcements of voluntary common stock issues and of -0.64% for common equity issuance driven by regulatory capital requirements. Cornett, Mehran, and Tehranian (1998) report an average CAR $(-1,0)$ of -1.62% for voluntary common stock issues and of -0.39% for involuntary issues. Krishnan et al. (2009) find an average CAR $(-1,0)$ ranging from -0.81% to -0.87% for equity issue announcements by banks.

Additional evidence that CPP injections did not convey an adverse signal comes from the banks that rejected CPP injections. If accepting CPP funds conveys negative information, the decision to reject CPP injections should convey positive news. Panel A of Table 5 shows, however, that announcements by banks declining participation in CPP and those rejecting CPP funds after receiving approval are both accompanied by statistically insignificant excess returns. This evidence is not consistent with CPP rejections being viewed as a signal of financial strength. Overall, the announcement returns do not support

⁹ These excess returns are slightly larger than those reported by Veronesi and Zingales (2010), but this difference is partially due to our exclusion of Wachovia, the wealth effects for which ought to be reflected in Wells Fargo's excess return due to its announced acquisition of Wachovia. Other minor differences between our event study approach and that in Veronesi and Zingales (2010) are that we use a narrower event window that excludes October 10, 2008, and that we use market model parameters to compute excess returns, whereas Veronesi and Zingales (2010) subtract beta-adjusted returns from raw returns.

¹⁰ One possibility is that when a firm is highly levered, the increase in firm value accrues entirely to bondholders. To explore this, we divide the sample of later recipients into those with high and low (nondeposit) leverage, but do not find meaningful differences between the excess returns for the two groups of banks.

either a positive certification effect or an adverse signaling effect associated with CPP injections.

4.2 Valuation effects of SCAP announcements

Given the more detailed evaluation of bank financial health conducted under SCAP, it is worth examining whether SCAP had a greater certification effect than CPP. To do so, we calculate excess stock returns around four dates when key details about SCAP were disclosed. SCAP was a component of the CAP initiative, which was first unveiled on February 10, 2009, as part of the Financial Stability Plan. In addition to the confounding effect of the other programs of the Financial Stability Plan that were announced on February 10, important details about SCAP were not disclosed on this date. There was also confusion about whether the program represented a move toward privatization of major banks. Consistent with this uncertainty, we find that excess returns on this date for the eighteen banks subject to CAP are sharply negative, as are excess returns for other public banks not subject to the program.¹¹

Important clarifications about the goals of CAP and the associated capital injections were provided on February 23 and 24. We find that excess returns for the eighteen banks subject to SCAP are sharply positive around these dates, averaging 17.1% (median of 16.4%). The methodology underlying the SCAP assessment and the assumptions regarding macroeconomic scenarios were disclosed on April 24, 2009. On this date, excess returns for the affected banks are indistinguishable from zero, but excess returns for banks exempt from the stress tests are significantly positive. This is consistent with a view that exempt banks also stood to benefit from a perceived reduction in systemic risk in the financial sector due to the SCAP assessments.

The results of the stress tests were partially leaked on May 6 and were officially announced on May 7, 2009. Excess returns for both affected and unaffected banks are significantly positive around these dates. Excess returns are sharply positive for the banks subject to SCAP and average 7.3% with a median of 5.2%, both of which are significant at the 1% level. Interestingly, the excess returns are also positive for the subsample of nine banks that were required to raise more capital following the SCAP assessment. In addition, there is evidence that the market partially anticipated that these banks would need more capital. When details about CAP were disclosed on February 23–24, the excess return for these nine banks averages 23.7% and is considerably higher than the average excess return of 10.6% for the banks that were later viewed to have sufficient capital. These large positive excess returns, particularly for banks that were required to raise new capital, are consistent with a strong certification role played by the SCAP process.

¹¹ Our analysis excludes GMAC, one of the nineteen institutions subject to SCAP, since it is not a publicly traded company.

5. Changing Political Environment and Costs of TARP Participation

Although signaling effects do not appear to play a role in bank CPP participation and rejection decisions, our evidence suggests that the compensation restrictions attached to the injections deterred participation and acceptance of funds. As discussed earlier, there was considerable controversy over compensation practices at participating banks. The sour political climate toward TARP recipients and their compensation practices is exemplified by H.R. 1586, which was proposed and passed by the U.S. Congress in under twenty-four hours with an overwhelming margin. This bill imposed a punitive tax of 90% on employee bonuses, retroactive to January 1, 2009, by institutions that had received \$5 billion or more in TARP capital. Some commentators believed that the bill would impose significant costs on banks by making it more difficult to retain key talent, and concerns over broader government involvement in TARP recipients were also raised.

We study stock prices of CPP recipient banks around the passage of H.R. 1586 on March 19, 2009, to examine whether the focus on compensation and potential government interference affected the costs of participation in TARP. Panel A of Table 6 shows that all of the eight financial institutions that were directly affected by the provisions of the bill experienced negative excess returns on that date, with an average excess return of -7.9% , which is statistically significant at the 1% level. In dollar terms, the loss in equity market capitalization for these eight banks was substantial, with a median decline of \$2.48 billion and an aggregate decline of \$24.3 billion using the estimated $(-1,0)$ excess returns.

In comparison, Panel B shows that excess returns around passage of the bill are essentially zero for other CPP recipients that received less than \$5 billion and hence were exempt under the bill. Panel C shows that banks that did not receive CPP capital experienced positive excess returns on March 19, 2009, averaging 1.4% , though the median excess return is zero. This non-negative valuation effect for nonparticipants suggests that the market did not expect these banks to be affected by the bill. Overall, the evidence is consistent with the view that the focus on compensation and the prospect of government influence on compensation were perceived to have costs for participating banks.

6. Repayment of CPP

If costs of participating in TARP increased since its inception, we expect that some banks would seek to exit the program. We expect that CEOs that bore the largest costs from the revised compensation rules announced in February 2009 are more likely to repay CPP injections. Banks may also choose to repay CPP injections if their ability to raise capital from public markets increased as financial markets stabilized. In fact, most banks were required by the U.S. Treasury to replenish all or part of the CPP capital

Table 6
Market reaction to H.R. 1586

Name	Obs.	Raw Return (-1;0), %	CAR (-1;0), %
Panel A: Affected TARP Recipients			
Bank of America		-9.6	-7.1
Citigroup		-15.6	-12.7
Goldman Sachs		-5.7	-3.4
JPMorgan Chase		-8.0	-5.7
Morgan Stanley		-13.0	-10.1
PNC Financial Services		-8.9	-7.2
U.S. Bancorp		-9.9	-8.2
Wells Fargo		-10.5	-8.2
Mean	8	-10.1***	-7.9***
Median	8	(-9.8***)	(-7.7***)
Panel B: Unaffected TARP Recipients			
Mean	186	-1.3**	-0.1
Median	186	(-1.3***)	(-0.4)
Panel C: TARP Nonrecipients			
Mean	194	0.5	1.4**
Median	194	(0.0)	(0.1)

This table presents raw and abnormal returns on March 19, 2009, when H.R. 1586 was passed by the U.S. House of Representatives. *Affected TARP Recipients* are banks that received more than \$5 billion under TARP. *Unaffected TARP Recipients* are banks that received less than \$5 billion under TARP. *CAR(-1;0), %* is the one-day abnormal return calculated using return on S&P 500 as market index and market model, with parameters estimated over the period from January 1, 2007, until September 17, 2008. Medians are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

that was repaid. In some cases, the announcement of new equity issuance occurs concurrently with repayment announcement, though many banks also raised equity in the days surrounding the repayment announcement without a concurrent announcement. The first repayments of CPP were made on March 31, 2009, by four banks, which expressed concerns over the dividend and compensation restrictions associated with TARP as well as the changing public and political perception of TARP.¹² As of November 30, 2009, ninety-five of our sample banks had announced their intention to repay CPP capital.

6.1 Likelihood of exit from CPP

Table 7 reports logit models where the dependent variable equals one if a bank has repaid CPP funds and zero if a bank has received CPP funds but has not repaid them.¹³ Models (1) and (2) display a positive coefficient on bank size, indicating that larger banks are more likely to have repaid CPP infusions. Repaying banks also have stronger capital ratios, as indicated by the positive coefficient on the Tier 1 ratio. Finally, asset quality is higher for repaying banks, as indicated by the negative coefficients on net charge-offs in model (1) and on loan loss provisions in model (2). Thus, banks repaying

¹² See "Four Small Banks Are the First to Repay TARP Funds," *New York Times*, March 31, 2009.

¹³ As an alternative to a logit model for the repayment decision, one could arguably estimate a hazard model for the time to repayment. This latter approach is, however, not feasible given publicly available data since we only observe when a bank has been granted permission to repay TARP by the U.S. Treasury and we cannot determine whether and when a bank sought permission to repay TARP.

Table 7
Bank CPP repayment decision

	(1)	(2)	(3)	(4)	(5)	(6)
Ln(Total Assets)	0.84*** (0.00)	0.82*** (0.00)	0.89*** (0.00)	0.85*** (0.00)	0.45 (0.15)	0.37 (0.22)
Derivatives/Assets, %	0.09 (0.25)	0.11 (0.15)	0.13** (0.04)	0.14** (0.03)	0.15** (0.02)	0.17*** (0.01)
Wholesale Debt, %	-6.19 (0.12)	-5.93 (0.14)	-4.85 (0.30)	-4.40 (0.32)	-4.51 (0.34)	-4.18 (0.36)
Tier 1 Ratio, %	0.31*** (0.01)	0.33*** (0.00)	0.24* (0.07)	0.25** (0.01)	0.27** (0.02)	0.28*** (0.00)
Net Charge-offs, %	-2.41*** (0.00)		-4.06*** (0.00)		-3.82*** (0.00)	
Loan Loss Provisions, %		-1.69*** (0.00)		-2.61*** (0.00)		-2.45*** (0.00)
Write-downs, %	-1.23 (0.25)	-0.39 (0.77)	-1.08 (0.46)	0.45 (0.79)	-0.52 (0.70)	0.53 (0.72)
Campaign Contributions, %	-21.53** (0.03)	-20.57** (0.02)	-23.24** (0.04)	-20.89* (0.05)	-17.31 (0.14)	-14.60 (0.19)
Fed Board	-1.33 (0.23)	-1.40 (0.23)	-1.30 (0.27)	-1.30 (0.29)	-1.07 (0.28)	-1.00 (0.32)
Excess CEO Compensation			1.74*** (0.01)	1.86*** (0.01)		
CEO Compensation > 500K					0.15** (0.02)	0.17** (0.01)
Real Estate / Total Loans, %	-0.08* (0.07)	-0.07 (0.11)	-0.09 (0.10)	-0.07 (0.19)	-0.12** (0.03)	-0.10** (0.04)
C&I / Total Loans, %	-0.13** (0.01)	-0.11** (0.03)	-0.16** (0.01)	-0.13** (0.03)	-0.19*** (0.00)	-0.16*** (0.01)
Consumer / Total Loans, %	0.01 (0.80)	0.02 (0.73)	-0.01 (0.82)	-0.01 (0.92)	-0.04 (0.47)	-0.04 (0.53)
State Macro Index Growth, %	0.03 (0.79)	0.02 (0.84)	0.00 (0.99)	0.01 (0.95)	-0.05 (0.69)	-0.04 (0.67)
Observations	305	305	245	245	245	245
Pseudo R-squared	0.34	0.35	0.41	0.42	0.40	0.41

This table presents logit model estimates of the bank decisions to repay CPP funds. The dependent variable is equal to one for TARP recipients that repaid CPP funds during the period from February 26, 2009, until November 30, 2009. It is equal to zero for other TARP recipients. Financial data are as of December 31, 2008. *Ln(Total Assets)* is the natural logarithm of total assets. *Derivatives / Assets, %* is the notional value of the bank's derivatives contracts scaled by assets. *Wholesale Debt, %* is the sum of subordinated notes, debentures, and other borrowed money scaled by assets. *Tier 1 Ratio, %* is Tier 1 capital scaled by risk-weighted assets. *Write-downs, %* are realized and mark-to-market losses on trading assets from January 1, 2008, to December 31, 2008, scaled by assets. *Net Charge-offs, %* is the amount of credit losses from January 1, 2008, to December 31, 2008, net of recoveries scaled by the average total loans. *Loan Loss Provisions, %* is the amount reserved from January 1, 2008, to December 31, 2008, to cover near-term expected loan losses scaled by the average total loans. *Campaign Contributions, %* are political contributions made by the bank-sponsored political action committee and the bank's employees to the members of the House Financial Services Committee in the 2008 election campaign. *Fed Board* is an indicator for the financial institutions that had executives on the boards of the Federal Reserve Banks in 2008 and 2009. *CEO Compensation > 500K* is the natural logarithm of the difference between the total CEO compensation and \$500,000 if it is positive and zero otherwise. *Excess CEO Compensation* is the residual from the regression of the logarithm of total CEO compensation in 2008 on the logarithm of the bank's total assets, return on assets, and age. *C&I / Total Loans, %* is the fraction of commercial and industrial loans in total loans. *State Macro Index Growth, %* is calculated as the weighted average of the changes in the Federal Reserve Bank of Philadelphia's state-coincident indexes from December 2007 to February 2009 with the share of deposits of a given bank in a given state used as weights. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. *p*-values based on heteroscedasticity-adjusted standard errors are reported in parentheses.

TARP funds have better-performing assets and are financially healthier than banks that continue to participate in CPP.

Models (1) and (2) also suggest that a bank's financial profile is not the only determinant of whether or not it repays TARP funds. Banks with a greater proportion of C&I loans are less likely to exit TARP, perhaps because of regulators' concern over the financial viability of banks that are heavily involved in lending to the corporate sector. In addition, we find that campaign contributions lower the probability of TARP exit. This suggests that banks that are politically well connected perceived the costs of remaining under TARP to be lower than banks that lacked strong political ties.

Models (3) and (4) add excess CEO compensation as an independent variable. The coefficient on this variable is positive and significant at the 1% level in both models. Thus, high levels of excess CEO compensation increase the probability that a bank exits TARP. This result is supportive of the view that CEOs who faced larger personal costs of increased government interference on executive compensation decisions were more likely to exit TARP. The economic importance of this variable is large. A one-standard-deviation increase in excess compensation leads to a 7.3-percentage-point higher probability of exit, a substantial increase relative to the 12.1% unconditional exit probability in the sample. Models (5) and (6) show that the log of the excess of CEO compensation over \$500,000 also has a statistically significant effect on TARP exit decisions. A one-standard-deviation increase in this variable is estimated to increase exit probability by 6.6 percentage points, or more than half the size of the unconditional exit probability.

The economic importance of the compensation variables appears to be as large as or larger than the other variables. A one-standard-deviation increase in size raises exit probability by 6.7 percentage points, and a similar increase in the Tier 1 ratio increases it by 1.6 percentage points, based on estimates in model (3). Increasing the net charge-offs ratio by a standard deviation is estimated to lower exit probability by 1.7 percentage points, and a similar increase in C&I loans is estimated to lower exit probability by 1.5 percentage points.

6.2 Announcement returns around CPP repayment

Panel A of Table 8 reports the raw and excess stock returns around the first public announcement that a bank is repaying CPP injections. Our sample consists of ninety-five repayment announcements, excluding ten banks whose stock did not trade on the announcement date. We find that the excess return averages 0.5% over the $(-1,0)$ interval with a median of 0%, neither of which is significantly different from zero. Dividing the sample into announcements accompanied with equity issuance and those without contemporaneous equity issuance news reveals a clear difference. Repayment announcements for the sixty-three banks uncontaminated with equity issuance news experience significantly positive excess returns, averaging 1.9%. However, announcement

Table 8
Market reaction to CPP repayments

Name	Obs.	Raw Return (−1;0), %	CAR (−1;0), %
Panel A: Announcements of CPP Repayments.			
All repayments			
Mean	95	0.5	0.5
Median	95	(−0.2)	(0.0)
No Stock Issue Ann. Same Day			
Mean	63	1.9*	1.9**
Median	63	(0.0)	(0.2*)
Stock Issue Ann. Same Day			
Mean	32	−2.1	−2.2**
Median	32	(−1.5*)	(−0.6**)
Panel B: Equity Research Analyst Expectation of Repayment.			
City National		−0.5	−0.9
Northern Trust		2.4	1.9
JPMorgan Chase		4.6	4
First Horizon National		3.6	3.1
Comerica		8.3	7.9
Bank of New York		9.6	9
Morgan Stanley		8.0	7.3
U.S. Bancorp		8.9	8.6
Mean	8	5.6***	5.1***
Median	8	(6.3***)	(5.7**)

Panel A presents abnormal returns on the day of the first public announcement of the CPP repayment with announcements made during the period from February 26, 2009, until November 30, 2009. Panel B presents abnormal returns on the day of issuance of a Goldman Sachs research report (March 11, 2009) naming eight banks most likely to repay CPP funds. $CAR(-1,0)$ is the one-day abnormal return calculated using return on S&P 500 as the market index and market model, with parameters estimated over the period from January 1, 2007, until September 17, 2008. *No Stock Issue Ann. Same Day* is a subsample of banks that did not announce an equity issue on the same day. *Stock Issue Ann. Same Day* is a subsample of banks that announced equity issue on the same day. Medians are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

returns are significantly negative for the thirty-two banks that contemporaneously disclose equity issuance, with excess returns averaging −2.2%.

These excess returns may underestimate the valuation impact of repaying CPP infusions since there was substantial media and equity analyst focus and speculation around the possibility of repayment by TARP banks. For example, in a highly publicized research report released on March 11, 2009, Goldman Sachs predicted that eight banks would repay TARP capital. Panel B of Table 8 shows that seven of these eight banks experienced positive excess returns on that day, and they are statistically significant with an average CAR (−1,0) of 5.1% and a median CAR (−1,0) of 5.7%.

Overall, the positive market reaction to the announcements of TARP repayment intent is consistent with the view that the cost of participating in TARP increased for banks since the program’s inception. It is also possible that the approval to repay TARP conveys positive information about the financial health of the banks, even though the initial capital injection does not appear to contain a certification effect.

7. Investor Expectations of Future Regulatory Action

We explore whether TARP altered investors' expectations regarding future regulatory interventions in the banking sector. Prior to CPP injections, the government had a senior claim on the bank's assets through the FDIC guarantee of bank deposits. This guarantee gave the government an incentive to protect the bank deposits, and to do so, regulators would require the bank to raise new capital or would seize the bank's assets when the value of deposits was in jeopardy. Since regulatory interventions and seizures wipe out much of the value of the common equity, investors assess the adequacy of a bank's capital in terms of the protection it offers against regulatory intervention. As deposits are senior to common and preferred equity and unsecured debt, the Tier 1 capital ratio serves as a measure of investor protection from regulatory intervention because it includes both common equity as well as debt components.

TARP potentially altered investors' perception of the risk of regulatory interventions and seizures because CPP infusions took the form of preferred equity and were junior to deposits. To protect taxpayers, regulators would now have to intervene sooner since common equity was the only buffer in the bank's capital junior to the taxpayer's claim. This implies that following CPP injections, the amount of common equity in the capital structure became more relevant to investors as a measure of protection against regulatory intervention.

These arguments suggest that CPP infusions may have had the unintended effect of increasing investor perception of the risk of regulatory seizures and the associated losses in shareholder value. They also suggest an explanation for why bank stock prices continued to decline after CPP was unveiled and the need for the SCAP that evaluated the adequacy of the common equity capital of banks. For example, from the announcement of CPP infusions on October 14, 2008, through May 7, 2009, the date when the SCAP results were announced, the KBW bank index dropped by 61%, compared with a 9% drop in the S&P 500 index over that period.

If CPP infusions made investors concerned about the adequacy of the common equity capital, we expect that share prices of banks should be correlated with the amount of common equity in bank capital following the CPP infusions. To test this proposition, we examine the cross-sectional variation in stock returns for all publicly traded U.S. banks between October 14, 2008, and May 7, 2009, the period between the introduction of CPP and the SCAP. For comparison, we also examine the pre-CPP period from July 1, 2007, to October 13, 2008, a timeframe that reflects the onset of the financial crisis for the banking sector. We calculate both cumulative stock returns and buy-and-hold abnormal stock returns (BHARs) over these intervals following the methodology in [Lyon, Barber, and Tsai \(1999\)](#), and we use these as the dependent variables in the regressions.¹⁴

¹⁴ Specifically, we calculate BHARs by forming 100 reference portfolios based on the deciles for market equity on October 13, 2008, and the book-to-market ratio at the end of the 2007 fiscal year for NYSE firms on CRSP. The

Our primary explanatory variable in these regressions is the amount of common equity in bank capital, which we calculate in two ways. First, we decompose the Tier 1 ratio into the Tier 1 Common ratio, which includes all the common equity components of Tier 1 capital, and the Other Tier 1 ratio, which includes all non-common equity components of Tier 1 capital. Second, we calculate the tangible common equity ratio, defined as the ratio of tangible equity to tangible assets. Unlike the Tier 1 Common ratio, the tangible common equity ratio removes the effect of goodwill and other intangible assets.

Table 9 reports regression estimates of bank stock returns during the financial crisis. We include a number of control variables that may be correlated with share price performance, including bank size, credit losses, funding mix, and loan composition. All financial ratios are calculated for the quarter immediately preceding the window used for calculating stock returns, with the exception of net charge-offs, which are calculated over the quarters most closely aligned with the stock return window.

Columns (1)–(4) display regression results for the post-CPP infusion period. We find that both raw returns and BHARs are positively correlated with the Tier 1 Common ratio as well as with the tangible common equity ratio. Thus, following CPP infusions, banks with higher ratios of Tier 1 Common and tangible common equity capital displayed stronger stock returns, consistent with the view that investors rewarded banks with more common equity capital. We do not find a similar effect for the Other Tier 1 ratio, which is estimated to be either negative or statistically insignificant, suggesting that non-equity-based capital was not viewed as favorably by investors during this period.

The positive association between stock returns and equity capital components in the post-CPP period is in contrast to the evidence from the pre-CPP period. Columns (5)–(8) report estimates from the pre-CPP period and show that the coefficient on the Tier 1 Common ratio is insignificant, while that on the tangible common equity ratio is negative. This suggests that bank stock returns became correlated with the common equity component of capital following CPP infusions, consistent with the view that the market perceived the risk of regulatory intervention to be higher following CPP infusions.

8. Conclusions

We study TARP, the largest financial rescue plan in U.S. history. We document that capital infusions were directed toward large banks and those with greater derivatives exposures, a pattern consistent with an objective of lowering systemic risk. Consistent with an objective of lowering distress costs, banks with less stable funding mixes were more likely to receive government

value-weighted average cumulative return for each portfolio is used as the expected return benchmark. We find a reference portfolio for each bank based on its market equity and book-to-market ratio and subtract the reference portfolio return from the bank's cumulative return.

Table 9
Bank stock returns before and after CPP capital infusions

	October 14, 2008–May 7, 2009				July 1, 2007–October 13, 2008			
	Raw Returns		BHARs		Raw Returns		BHARs	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Tier 1 Common Ratio, %	0.65*** (0.01)		0.69*** (0.00)		0.11 (0.61)		0.12 (0.57)	
TCE Ratio, %		0.62** (0.04)		0.72** (0.02)		−0.57** (0.02)		−0.51** (0.04)
Tier 1 Other	−1.57*** (0.00)	−2.11*** (0.00)	−1.70*** (0.00)	−2.27*** (0.00)	−0.54 (0.65)	−1.31 (0.26)	−0.10 (0.94)	−0.82 (0.48)
Ln(Total Assets)	1.38* (0.07)	1.29* (0.10)	1.91 ** (0.02)	1.86 * (0.02)	2.74*** (0.00)	1.83* (0.06)	1.79* (0.06)	0.93 (0.33)
Net Charge-offs, %	−14.54*** (0.00)	−14.80*** (0.00)	−15.03*** (0.00)	−15.32*** (0.00)	−12.06*** (0.00)	−12.23*** (0.00)	−11.21*** (0.00)	−11.38 * ** (0.00)
Wholesale Debt, %	0.02 (0.88)	0.01 (0.96)	−0.03 (0.87)	−0.04 (0.82)	−0.62*** (0.01)	−0.69*** (0.00)	−0.61*** (0.01)	−0.67 * ** (0.00)
Real Estate / Total Loans, %	−0.14 (0.43)	−0.19 (0.26)	−0.19 (0.28)	−0.25 (0.16)	−0.45* (0.08)	−0.45* (0.07)	−0.34 (0.19)	−0.34 (0.18)
C&I / Total Loans, %	−0.31 (0.12)	−0.38 * (0.05)	−0.39 * (0.05)	−0.48 * (0.01)	−0.51* (0.07)	−0.47* (0.08)	−0.43 (0.12)	−0.39 (0.15)
Consumer / Total Loans, %	0.17 (0.40)	0.13 (0.52)	0.09 (0.68)	0.05 (0.82)	0.35 (0.36)	0.37 (0.31)	0.57 (0.13)	0.60 (0.10)
Observations	500	500	500	500	477	477	477	477
Adjusted R-squared	0.23	0.22	0.23	0.22	0.12	0.12	0.10	0.10

The dependent variables in the OLS regressions in this table are cumulative raw and buy-and-hold abnormal returns (BHARs) for all U.S. public banks over the period before and after the announcement of CPP capital infusions (October 14, 2008). Bank financial data are as of the latest quarter before each time period. *Tier 1 Common Equity Ratio*, % is Tier 1 common capital scaled by risk-weighted assets. *Tangible Common Equity Ratio*, % is common equity divided by total assets after reducing both amounts by goodwill, other intangible assets, and related deferred taxes. *Tier 1 Other*, % is Tier 1 capital minus Tier 1 common capital scaled by risk-weighted assets. *Ln(Total Assets)* is the natural logarithm of total assets. *Net Charge-offs*, % is the amount of credit losses over each time period net of recoveries scaled by the average total loans. *Wholesale Debt*, % is the sum of subordinated notes, debentures, and other borrowed money scaled by the bank's total assets. *C&I / Total Loans*, % is the fraction of the commercial and industrial loans in the total loans. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. *p*-values based on heteroscedasticity-adjusted standard errors are reported in parentheses.

infusions. Importantly, CPP infusions did not support banks with distressed loan portfolios.

CPP infusions were viewed to be costly by banks, and some banks opted out of participating in the program or rejected the capital infusions. We show that the healthier banks voluntarily turned down TARP assistance. Though the negative signal associated with a capital infusion is often considered to be an obstacle in enlisting bank support for a capital infusion program, we do not find evidence that CPP infusions were associated with an adverse signaling effect. Instead, our results suggest that government interference in executive compensation decisions was a major concern for banks, leading many of them to reject TARP support and deciding to exit TARP. Consistent with the importance of executive compensation scrutiny at TARP recipients, we show that the passage of the AIG bonus tax bill lowered the valuations of affected TARP recipients, while banks that did not receive TARP support experienced an increase in valuation. We show that banks where CEOs earned high levels of compensation were more likely to repay TARP funds and that TARP repayments were associated with significantly positive announcement returns. Our results point to a key role that executive compensation considerations played in the implementation of the capital infusion program.

Our results also suggest a potential unintended consequence of CPP infusions. Though they were designed to be more acceptable to banks by virtue of being nondilutive, the preferred equity claim held by the government shifted investor focus toward the common equity component of bank capital, which was unchanged by CPP infusions. As a result, CPP infusions did not stem the losses in bank valuation until the implementation of the SCAP stress tests, which assessed the adequacy of common equity and mandated banks to replenish it if necessary.

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